Add the directions before the lengths

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Resultant / ES: resultante / PT: resultante

Sum: first tail to final tip.

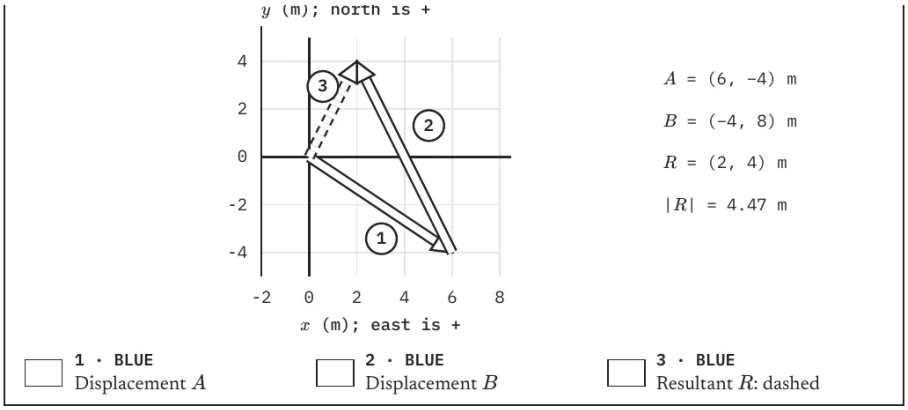
Also on your note page: Component, Magnitude, Quadrant.

B READ THE RELATIONSHIP

Add signed components along each axis

As you read, find: horizontal · vertical · resultant.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Two displacements are A = (6, −4) m and B = (−4, 8) m. Find their sum's magnitude and direction.

Given + x is east + y is north A and B use these axes.

Rule Add x and y . Use the right triangle.

Work R = (2, 4) m | R | = √ 20 m θ = $\tan^{-1}(4/2)$

Answer 4.47 m 63.4° north of east .

E SAVE FOR THE FINAL MEETING

XJ's motion must be tracked in more than one direction. Adding signed components keeps sideways motion in the forecast instead of treating every displacement as motion toward Earth.

Read the physics: Resultant

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

A vector has a size and a direction. Draw the next arrow with its tail at the first arrow's tip. Do not turn the arrow when you move it.

02 HOW IT WORKS

Draw the resultant from the original start to the final tip. To calculate, add horizontal parts together. Add vertical parts separately. The two totals make a right triangle.

03 WHAT TO CHECK

Use the sides to find the length and the signs to choose the direction. State the axis you measured the angle from.

04 TRY THE EXAMPLE FROM YOUR NOTES

Two displacements are A = (6, −4) m and B = (−4, 8) m. Find their sum's magnitude and direction. 4.47 m 63.4° north of east .

Model check: Add signed components along each axis.

05 CONNECT TO XJ

XJ's motion must be tracked in more than one direction. Adding signed components keeps sideways motion in the forecast instead of treating every displacement as motion toward Earth.

SAY IT BEFORE YOU WRITE

Where does the resultant start and end? The two parts are ___ and ___. Together they point ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Component

ES: componente / PT: componente

Meaning in my words	How can you tell?
How would you explain component to a friend? In my words, this means ___.	If east and north are positive, which component is negative for a southeast trip? I can tell because ___.
Example or sketch	**Non-example and why**
What two arrows could you use to build a displacement of 3 m east and 4 m north? My example shows ___.	A classmate counts the diagonal and both parts as three separate trips. What is counted twice? I would change ___ because ___.

Resultant

ES: resultante / PT: resultante

Meaning in my words	How can you tell?
How would you explain resultant to a friend? In my words, this means ___.	Where does the single arrow start and end on a tip-to-tail drawing? I can tell because ___.
Example or sketch	**Non-example and why**
Walk 3 m east, then 4 m north. What single arrow joins your start and finish? My example shows ___.	Someone labels the resultant 7 m. Which length did they add up? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Magnitude

ES: magnitud / PT: módulo

Meaning in my words	How can you tell?
How would you explain magnitude to a friend? In my words, this means ___.	For a displacement of 5 m west, what part gives the magnitude? I can tell because ___.
Example or sketch	**Non-example and why**
A vector has perpendicular parts of 3 m and 4 m. How long is the diagonal? My example shows ___.	A partner gives the magnitude as −5 m. What is the minus sign doing? I would change ___ because ___.

Quadrant

ES: cuadrante / PT: quadrante

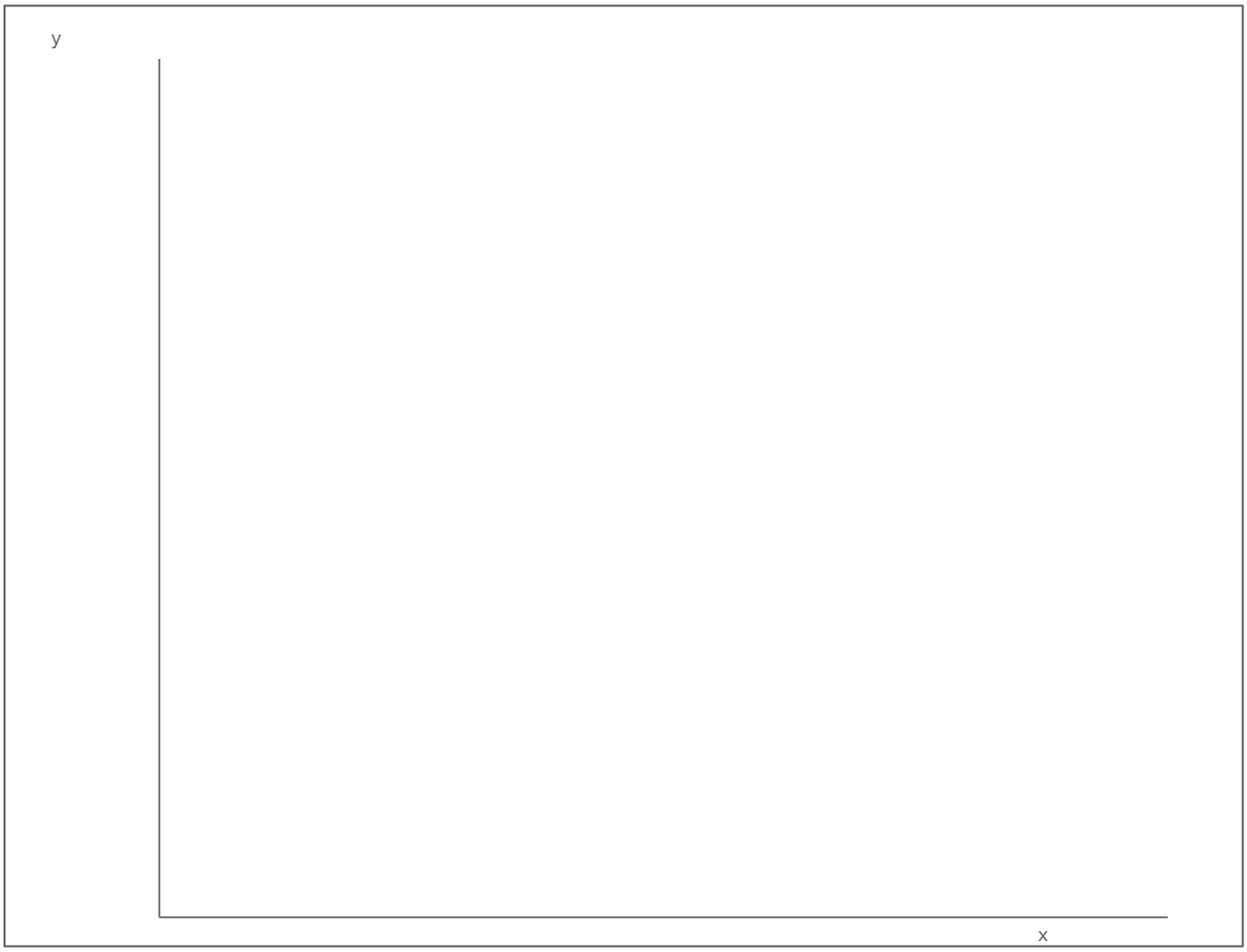
Meaning in my words	How can you tell?
How would you explain quadrant to a friend? In my words, this means ___.	Which way does an arrow point if its horizontal part is negative and its vertical part positive? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch a vector with components +2 m and −3 m. Where is its tip relative to its tail? My example shows ___.	Do two vectors with the same length always point into the same quadrant? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a right-angle journey; label x, y, and the resultant with its direction.

Labels you can use: horizontal · vertical · resultant.



Tell a partner

The two parts are ___ and ___. Together they point ___.

Send the whole movement

06

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Combine two displacement components into one arrow.

CHOOSE ONE JOB TOGETHER

NAVIGATION ENGINEER ☐

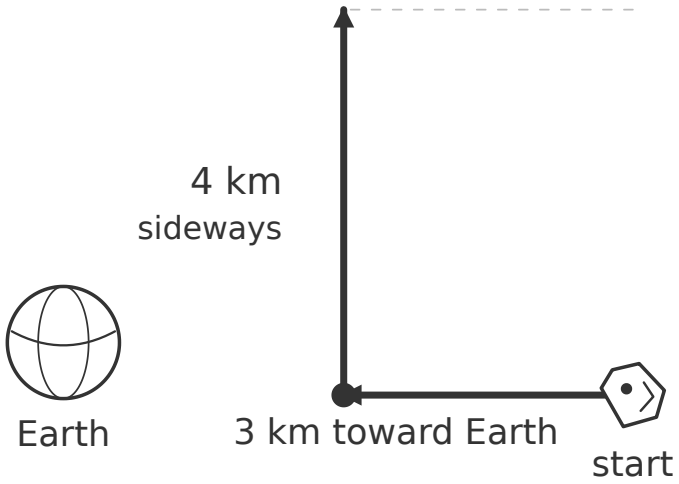
Draw one arrow for the whole displacement. Give its length and direction, and label both component legs.

MAP PRESENTER ☐

Use the map to report the whole movement. Explain why '**7 km** closer to Earth' is the wrong caption.

Word help: component / sideways / resultant

A tracker sees a **3 km** displacement toward Earth and **4 km** sideways. The second telescope needs the whole movement.



01 OUR FIRST CALL

What full movement should the second telescope track? Use your map.

The whole displacement is ___ toward ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What part can the updated tracker measure? What part will it miss?

Now we would say ___ because ___.

Switch speakers. Give the direction as an angle. Name the line you measure that angle from.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks